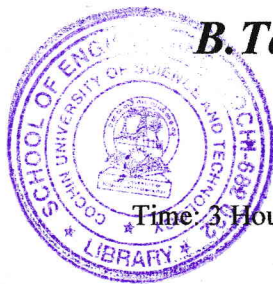


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B.Tech. Degree IV Semester Examination April 2019

CS/IT 1405 DATA STRUCTURES AND ALGORITHMS (2012 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Write binary search algorithm and explain its complexity.
- (b) Explain different hashing functions.
- (c) What is a stack? Explain basic stack operations.
- (d) Write an algorithm to evaluate postfix expression.
- (e) What is complete binary tree? Explain threaded binary tree using a diagram.
- (f) What is binary search tree? Explain traversal methods in a binary search tree.
- (g) What is a graph? Explain different graph representation methods using examples.
- (h) Compare and contrast BFS and DFS.

PART B

(4 × 15 = 60)

- II. Write quick sort algorithm and explain its average case complexity. (15)
- OR
- III. Write merge sort algorithm and explain its average case complexity. (15)
- IV. (a) Write an algorithm to convert an infix expression to postfix. (10)
- (b) Convert the infix expression $A + B * (C - D)$ to postfix. (5)
- OR
- V. Explain insertion, deletion, searching and traversal of a singly linked list. (15)
- VI. Explain insertion, searching and deletion algorithms for a binary search tree. (15)
- OR
- VII. What is an AVL tree? Explain insert and delete algorithms for an AVL tree. (15)
- VIII. Explain the Dijkstra's algorithm to find the shortest path with an example. (15)
- OR
- IX. (a) Write Prim's algorithm for constructing a minimum spanning tree. (8)
- (b) Write short notes on: (7)
 - (i) B Trees
 - (ii) B+ trees
